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India Smart Utility Week

ISGF – India Smart Grid Forum

INDIA – BRAZIL SMART ENERGY WORKSHOP

12 March 2026

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ASOCIACIÓN
DE DISTRIBUIDORAS
DE ENERGÍA ELÉCTRICA
LATINOAMERICANAS

Adelat's Members

31	10	+72M	+349	164M
Members	Countries	Consumers	TWh Total	People

Argentina	Bolivia	Brasil	Chile	Colombia	Costa Rica	Ecuador	El Salvador	Guatemala	Perú
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Edea	elfec	enel São Paulo	enel Chile	ESSA Grupo epm				eegsa Grupo epm	cvc energía
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Engineering Education

Strategic Pillars

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Knowledge Generation

- Regulatory Paper
- Investment Paper
- DSO Briefs

Build Alliances

- Multilaterals
- Associations
- Development Banks
- Think Tanks
- Academy

Insert into the Public Agenda

- Interaction with authorities
- Industry forums
- Opinion columns
- Media publications and interviews

Corporate Governance

Building Shared Knowledge

Regulatory Challenges and Improvements for Electric Distribution in the Latin American Energy Transition

The renewal of power distribution concessions – Analysis of the Brazilian case from the perspective of Latin American countries

The quality of power distribution in the Energy Transition: challenges and best regulatory practices

Distributed Generation: Towards a Sustainable Regulatory Model for Electricity Distribution

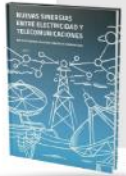
The resilience of power distribution systems: recommendations for a new regulatory paradigm

No investment, no transition: The future of power distribution in Latin America

Digitalization and Automation of power distribution: drivers and international good practices

Accelerating Smart Metering: Regulatory Drivers for Deployment in Latin America

The **distribution grid** as an enabler of **electric mobility** in Latin America

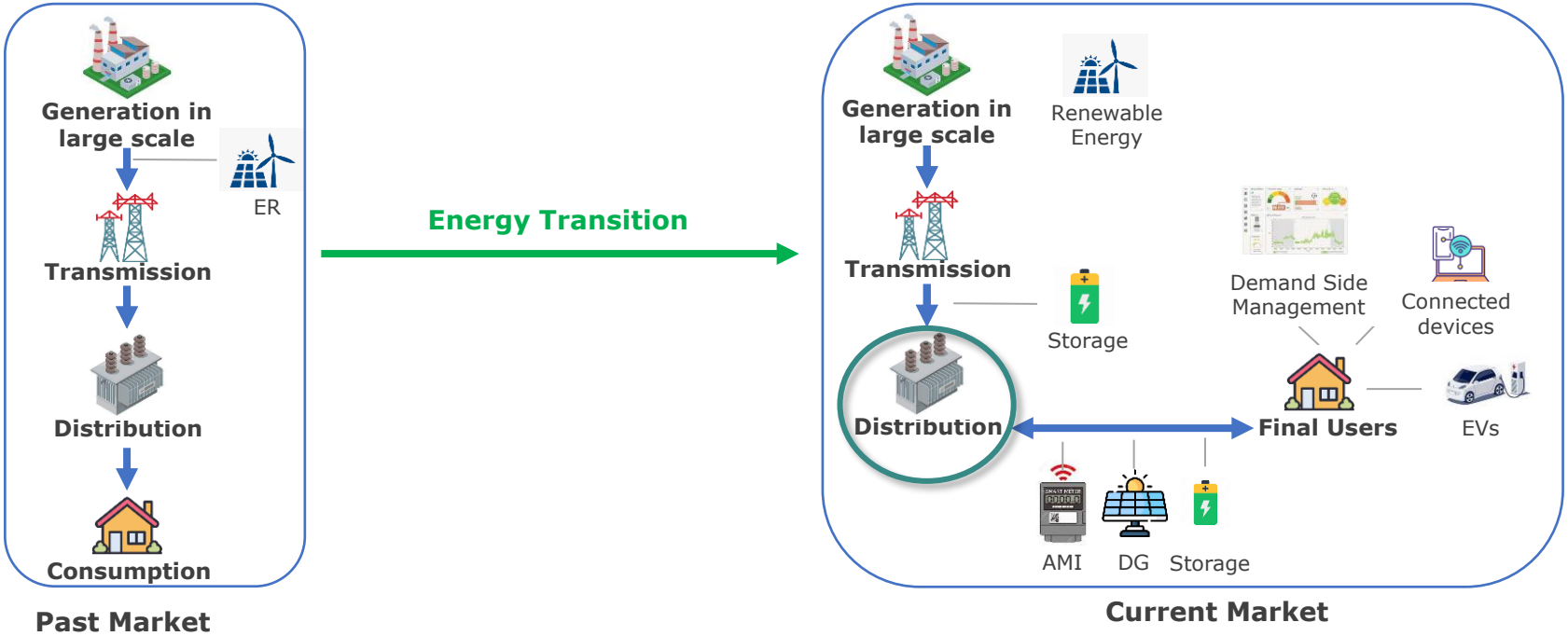


Digital Companies of the Future: New Frontiers in Shared Services for Electric Power and Telecommunications

Liberalization of retail markets in Latin America: new scenarios for electricity distribution

Access our papers

The new role of electrical distributors



Macro-challenges of distribution

Regulatory Challenges and Improvements for Electric Distribution in the Latin American Energy Transition

1. Increase quality and resilience

- Introduce the concept of **resilience**
- **Digitize and automate** the grid
- **Expand** the distribution network mesh
- Extend **smart metering**
- Ensure **coordination** of DERs

2. Sustainably transform the network

- Facilitate **new technologies, actors and business models** entrance
- Increase network **flexibility**
- Encourage **regulatory sandboxes**
- Strengthen **cybersecurity**

3. Foster economic efficiency

- Apply **hourly tariffs** and encourage demand response mechanisms
- Efficiently incorporate **Advanced Metering Infrastructure** (AMI)

AMI: Enabling Platform for the New Distribution Model

Observability, flexibility, and data management at the edge (edge computing)

Evolution

- From basic telemetry to end-to-end digital infrastructure with bidirectional communications and control.

Functions

- Granular data to identify losses, anticipate congestion, and support FLISR (Fault location, isolation, and service restoration).

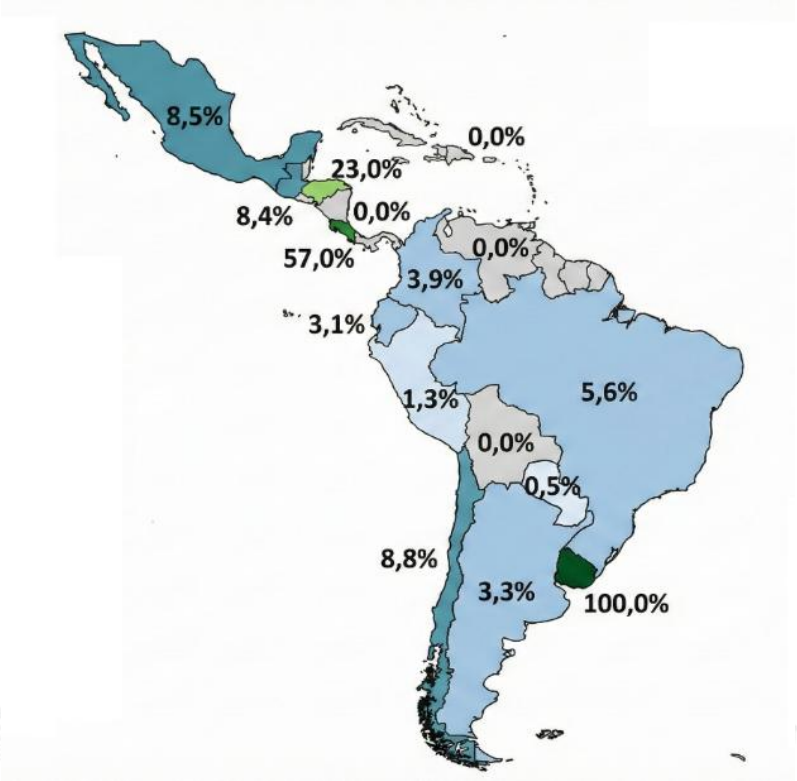
Integration

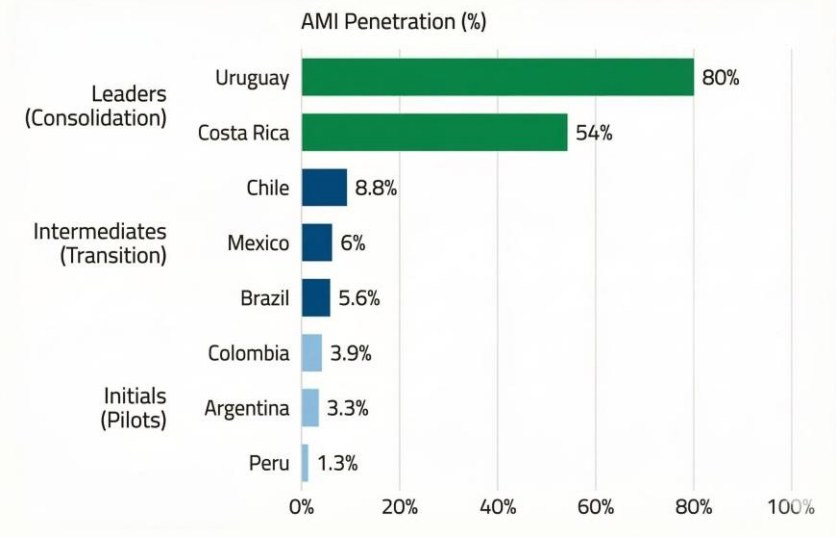
- Bidirectional flow management to integrate PV generation, storage and new profiles.

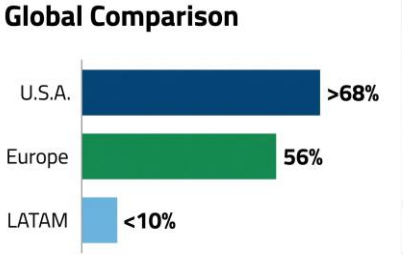
User

- Enable hourly tariffs, demand response, and smart EV charging.

With few exceptions, LATAM is in the early stage of AMI deployment







Implementation gap

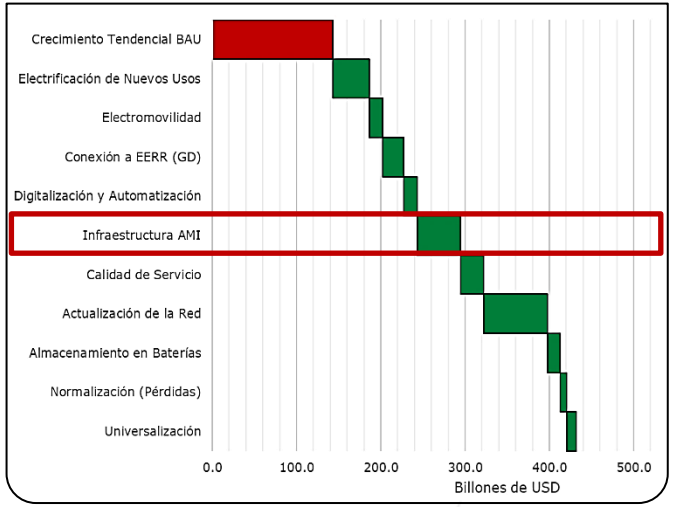
While global leaders renew towards AMI 2.0, the majority of LATAM remains under 10% coverage.

Progress depends on state mandates and isolated initiatives.

Investment required in AMI for the energy transition

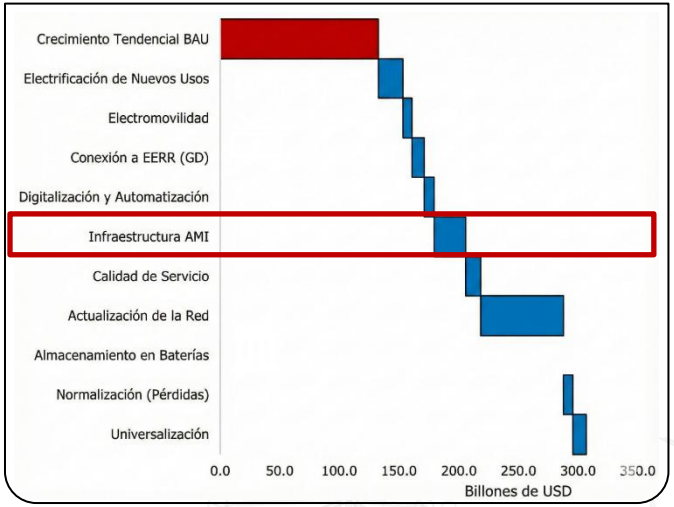
Effective ET 2040

100% AMI
USD 51.4 bn



Parcial ET 2040

50% AMI
USD 25.7 bn



Deployment strategies in the region

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National goals (Top-Down)

- **Costa Rica**: 100% by 2028 (ENREI strategy)
- **Colombia**: 75% by 2030 (conditional on positive CBA)

Technical-Focused Approach

- **Brasil**: proposal for a mandate of 2% additional per year.
- **Perú**: planning focused on large consumers

The Chilean case (lesson learned)

- **Chile**: from compulsory to voluntary after social rejection
- Consequence: stalled deployment, limited to failure replacement

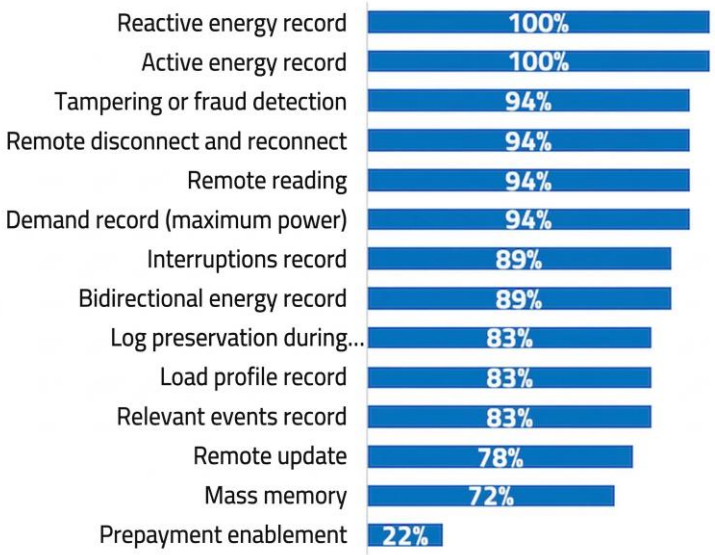
'De facto' strategy (Bottom-Up)

- **Argentina/Guatemala**: Isolated corporate initiatives for loss control.
- No national roadmaps

The lack of roadmaps in many countries limits deployment to 'niche profitability' (large users or fraud zones), preventing large-scale rollout.

Features: "Leapfrog" Opportunity to AMI 2.0

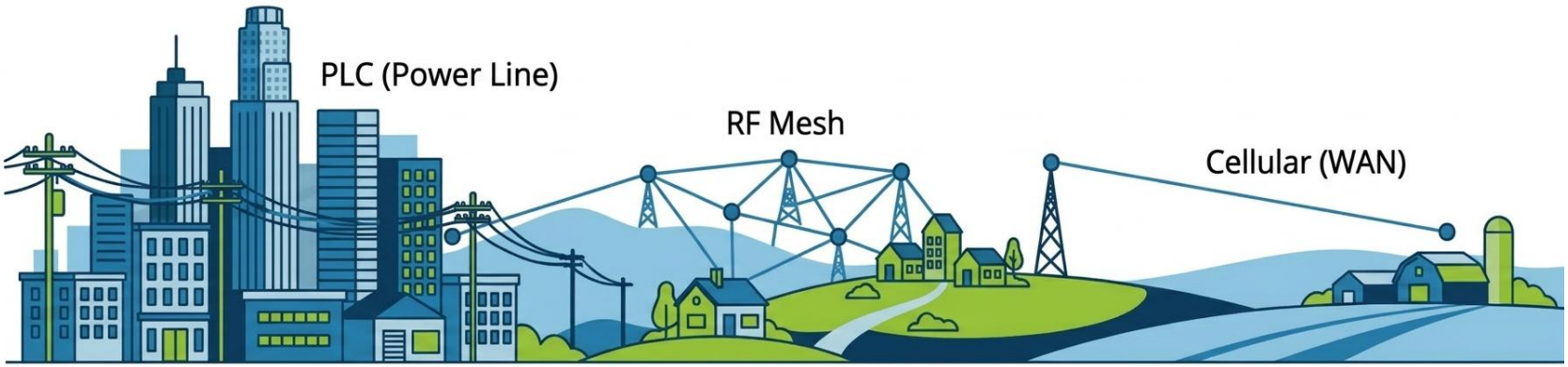
AMI's features reported by distributors



The region has the opportunity to skip stage 1.0 and adopt devices with Edge Computing capability, to avoid early obsolescence.

Communications: context LATAM demands hybrid architectures

There is no such thing as a "silver bullet": The Latin American reality demands the simultaneous management of hybrid networks



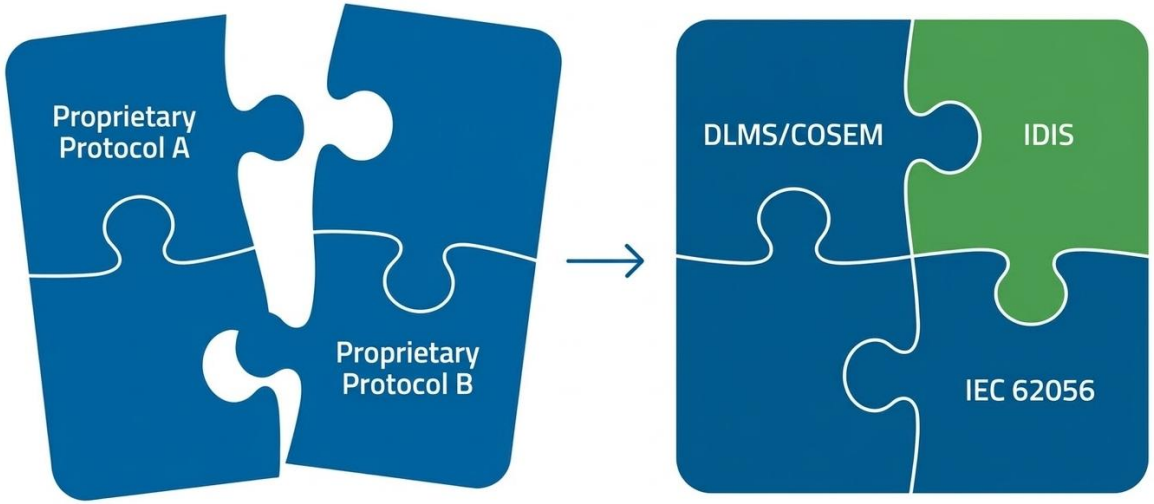
Pros: Cost.
Cons: Electrical noise, illegal networks.

Pros: Redundancy.
Cons: Requires high density.

Pros: Immediate coverage.
Critical Risk: Obsolescence (2G/3G Sunset).

Risk of obsolescence:
The shutdown of 2G/3G networks threatens investments if a coordinated technological transition is not planned

Interoperability: avoiding the "Tower of Babel"



Risk: Vendor Lock-in and high integration costs.

Solution: Open Standards.

The problem

The meter is useless if the data does not flow. The predominance of **closed solutions** generates vendor Lock-in

Current Solution

High integration with **billing**, low integration with **operations**

Recommendation

National Interoperability Standards and Independent Certification Laboratories

Integration and analytics: turning data into value

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Corporate Systems

- Predominant integration with business and billing systems. Reduced integration with SCADA/OMS.

Mechanisms

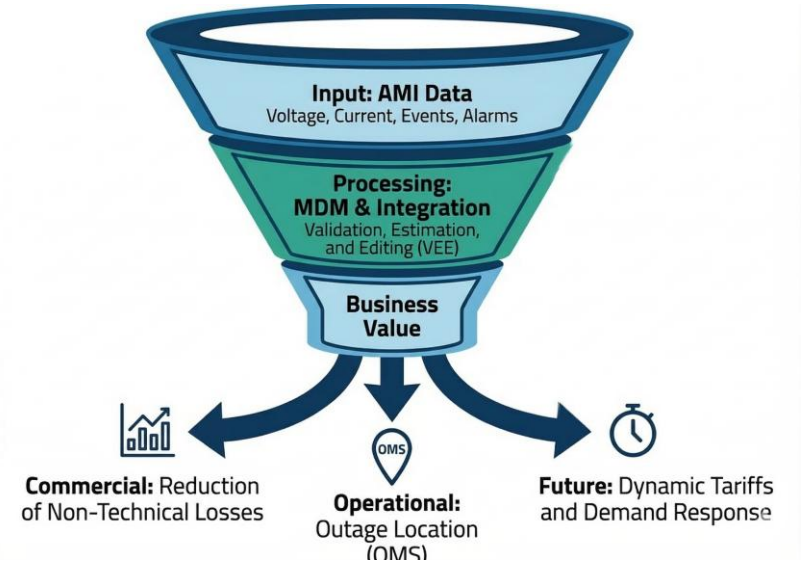
- Use of APIs, web services, and integration buses. Move towards modular schemes.

Analytics

- Focus on operational efficiency: detection of losses, validation of consumption and balances.

Potential

- Incipient exploration of operational applications for greater observability and proactivity.



Uneven integration Most integrate AMI with billing, but few have integration with SCADA/OMS for Network Automation (FLISR).

Information governance and user role

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Ciberseguridad

Expanded attack surface. Need for **ISO 27001** standards and end-to-end **encryption**

Data Privacy

Legal vacuum in the region. The regulation is usually **generic** (Habeas Data) and **not specific** to energy.

Social License (the User)

Transparency: Access to information is key. If the user perceives costs without seeing the advantages, rejection is inevitable (case of Chile).

Trust is the most critical asset for massification

Ownership and financing models

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Explicit Recognition
Brazil/Costa Rica

Property of the distributor transferred to tariff.

Challenge: Regulatory useful life (30 years) vs. technological useful life (15 years).

Transition / Conflict
Chile/Colombia

The problem of 'Stranded Assets'.

Who pays for the old unamortized meter that is retired?

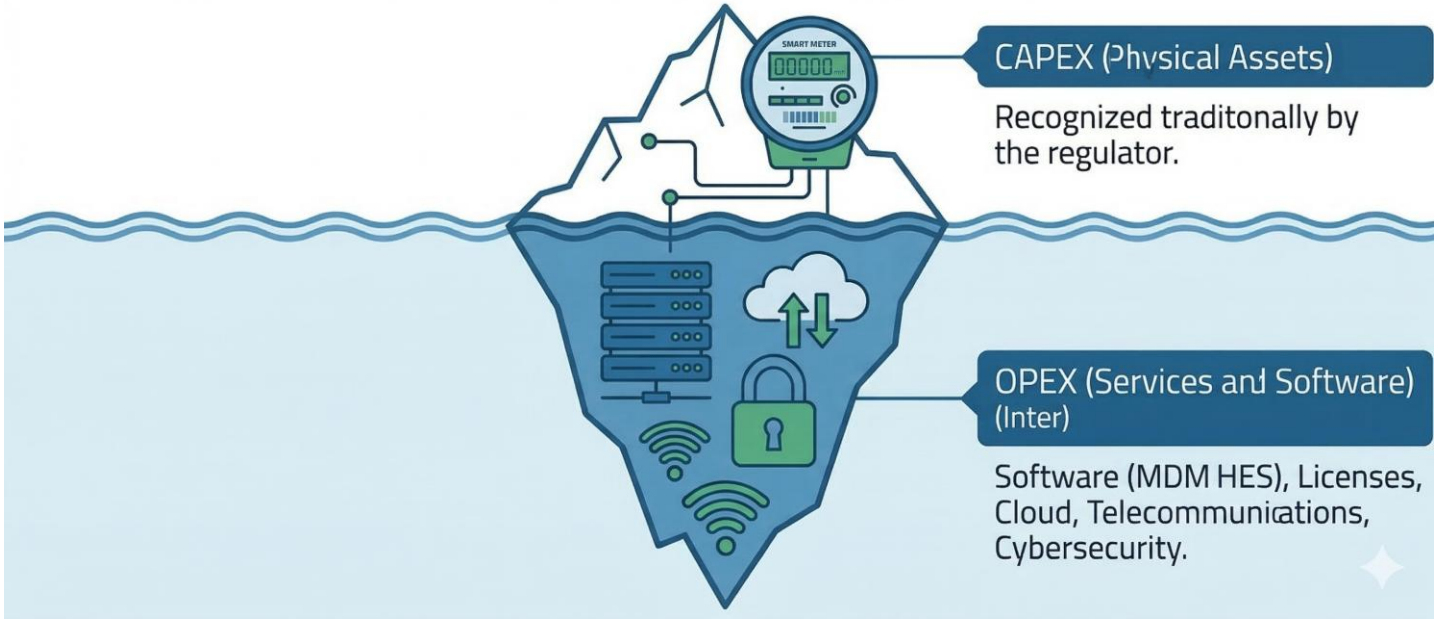
No Recognition
Argentina/Peru

Investment at company risk.

Only viable in areas of high fraud.

Critical Barrier:
Lack of accelerated depreciation for digital assets (software)

The Economic Equation: The CAPEX vs. OPEX Challenge



Need for TOTEX models

Regulation must recognize the full cost of the digital solution. Without remuneration for software and services, AMI infrastructure is unviable

The AMI Roadmap: Four regulatory pillars

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Progressive deployment

Prioritize value segments (large users, high-loss areas) before universalization.

TOTEX Recognition

Remunerate the entire ecosystem (hardware + software) and resolve stranded assets.

Social Cost-Benefit

Value externalities: resilience, quality and decarbonisation

Technological neutrality

Requiring open standards (interoperability) without prescribing communication technology

Conclusions and recommendations: Inaction is costly

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The technology is ready, but regulation must evolve and create incentives

'Leapfrog': LATAM can skip stages to AMI 2.0

Turning the current lag into an opportunity for modernization based on hybrid models

Ensuring interoperability by design

Putting the user and trust at the heart of the strategy

AMI is not an option, but a condition for transition

"We work together to promote collaboration and leadership in the energy transition in Latin America"

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